

the engineering field. Getting such large computational power to resolve business problems was expensive. The 'custom-supercomputer' soon gave way to the 'commodity-based supercomputer', popularly called HPC clusters, which are built from standard commodity off-the-shelf servers (mostly x86) and high-speed interconnects. HPC clusters provided some relief to cash-strapped research organisations in India. Yet, achieving high computing density, flexibility and smooth migration were common hurdles that most HPC solution providers in India wouldn't overcome.

Getting the Digital Wave advantage

Digital Waves began its odyssey in the Indian supercomputer field with Power-X HPC Clusters. These offered a basic HPC environment on the latest hardware, based on customer specifications and budget. The typical Power-X HPC Cluster included: dual processor-based servers (8, 16, 24 or 32 numbers), 2x Quad-Core Intel Xeon/Hexa-Core AMD Opteron processors, 8 or 16 GB RAM, 250 GB SATA HDD, a 1U/2U rack-mount server chassis, Gigabit/Infiniband interconnects and the Rocks Cluster Suite. These HPC clusters were for number crunching and research in physics, chemistry, bio-informatics, space research, weather modelling, CFD (computational fluid dynamics), oil and gas exploration, digital content creation (DCC), etc.

"We built Asia's first AMD Opteron-based cluster (render farm) for DCC [Digital Content Creation] in 2003," says Kaushik Sonawala, managing director, Digital Waves. Ranganath adds, "We use TYAN Server products and

integrate our own brand of Power-X HPC Clusters. We offer 2P, 4P and high-density rack-mount servers. Being a solutions partner with Mellanox, we offer Infiniband interconnects also."

Today the firm's customer base includes the who's who among research/academic institutions. Digital Waves is currently working with several research organisations like the Indian Institute of Science (IISc), Raman Research Institute (RRI), the IITs, Tata Technologies Limited (TTL), CSIR Labs, several defence/space research organisations, CDAC, GE, etc. They are also approaching government organisations that need high-density blade servers.

Power-X punch

Digital Waves' Power-X HPC Clusters address the common problems of cost, scalability and migration. "Any MPI (Open MPI or MPI-CH) based parallel application for scientific number crunching or computational research can run on our HPC clusters. These are standard applications, which can be easily migrated to from a stand-alone server," says Ranganath. "Most of our Power-X HPC clusters are installed with open source software. We use the Rocks Cluster Suite. The basic codes/applications are tested and run mostly on the open source platform, making migration on the HPC cluster very easy," elaborates Ranganath. Further, open source software helps Digital Waves give its customers greater efficiency, security and a price advantage.

Power-X HPC Clusters also provide greater flexibility, as the solutions are scalable. "For applications that can scale linearly, our HPC clusters can be scaled accordingly.



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